EV Charging Infrastructure Intelligence Dashboard

Power BI Data Analytics Project

Presented By –

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Project Overview

From raw EV station data to an interactive BI dashboard

Purpose

- Analyze EV charging infrastructure across countries, regions and operators.
- Clean and standardize raw station data using Power Query.
- Create calculated columns and analytical measures using DAX.
- Present results through three interactive Power BI pages.
- Generate insights for infrastructure and operational analysis.

Workflow

- Raw Dataset
- Data Inspection & Cleaning
- Power Query Transformation
- Calculated Columns & DAX Measures
- Dashboard Development
- Insights & Conclusion

Dataset & Data Preparation

Key steps used to make the dataset analysis-ready

Dataset

- 10,000 original station records
- 14 original columns
- Station, location, operator, status, connector and date information
- Multiple countries and geographic regions

Cleaning & Transformation

- Missing-value handling
- Duplicate checking
- Column renaming
- Text trimming
- Country-name standardization
- Operator and status standardization
- Connector-type transformation
- Latitude/longitude validation
- Charging capacity categories

Power Query, M Language & DAX

The transformation and calculation layer

Power Query / M Language

- Missing-value handling
- Duplicate checks
- Text cleaning
- Country mapping
- Operator cleaning
- Status standardization
- Connector transformation
- Geographic validation
- Capacity categorization

Calculated Columns & Measures

- Station Age
- Connector Category
- Operational Flag
- Connector Capacity Group
- Country Region
- Total Stations
- Operational Stations
- Non Operational Stations
- Total Connectors
- Average Connectors per Station
- Operational %
- Top Operator

Dashboard Structure

Three pages for overview, comparison and detailed investigation

01 Executive Overview

- KPIs, country distribution, map, operational status, monthly connectors and capacity.
- Interactive navigation
- Filters / slicers
- Detailed investigation

02 Infrastructure Analysis

- Operators, regions, connector types, capacity and country/status comparison.
- Interactive navigation
- Filters / slicers
- Detailed investigation

03 Station Details

- Station-level records, capacity distribution and regional operational summary.
- Interactive navigation
- Filters / slicers
- Detailed investigation

Page 1: Executive Overview

High-level view of the EV charging infrastructure

10K	**9K**	**626**	**93.46%**	**14K**
Total Stations	Operational Stations	Non Operational Stations	Operational %	Total Connectors

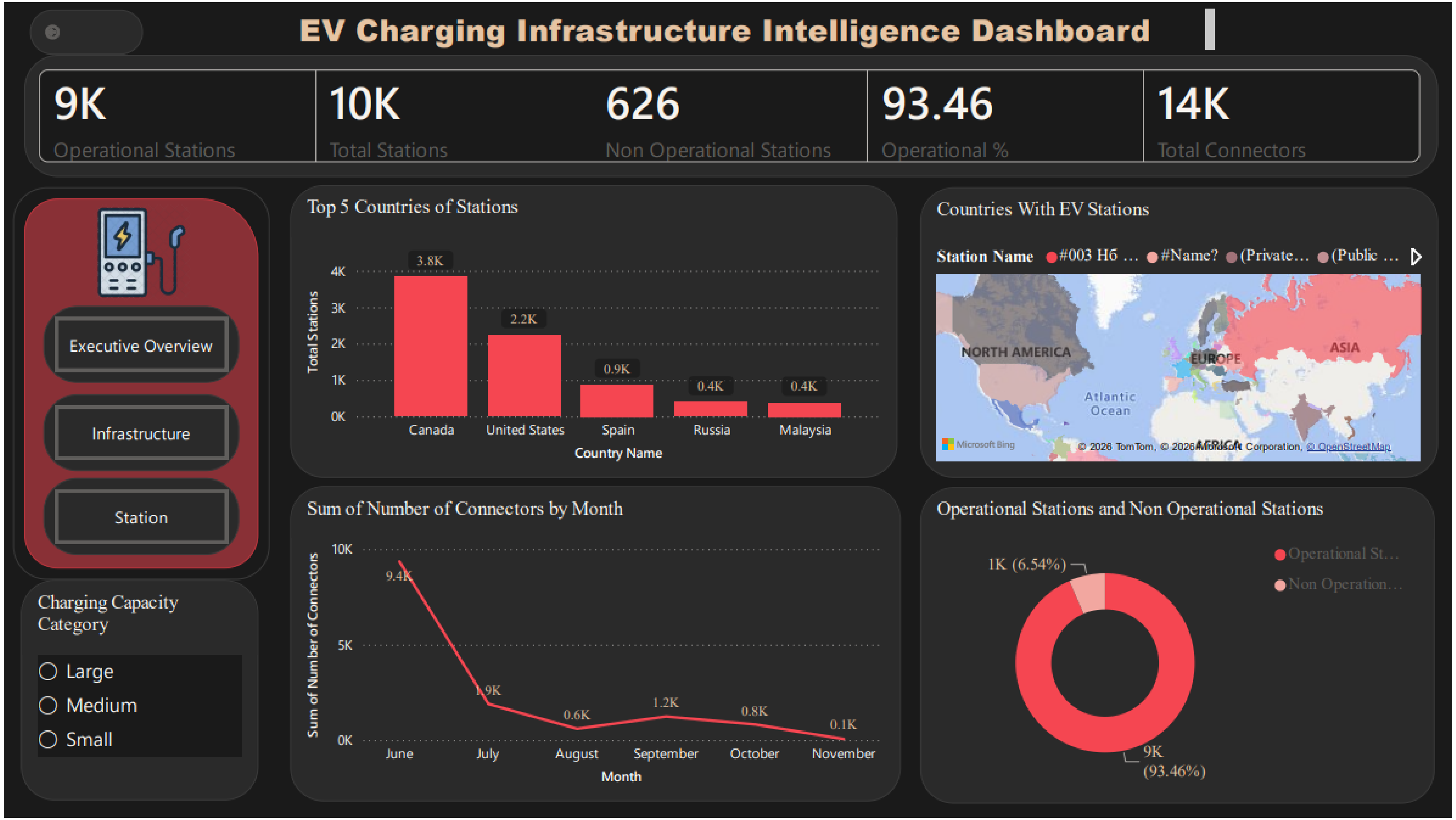
Main Visuals

- Top 5 Countries
- Countries With EV Stations map
- Operational vs Non Operational
- Connectors by Month
- Charging Capacity Category

Insights

- Canada: about 3.8K stations
- United States: about 2.2K
- Operational share: 93.46%
- June has about 9.4K connectors in the monthly chart
- Geographic distribution is concentrated in selected regions

Page 1: Executive Overview



Page 2: Infrastructure Analysis

Comparing operators, regions, connectors and capacity

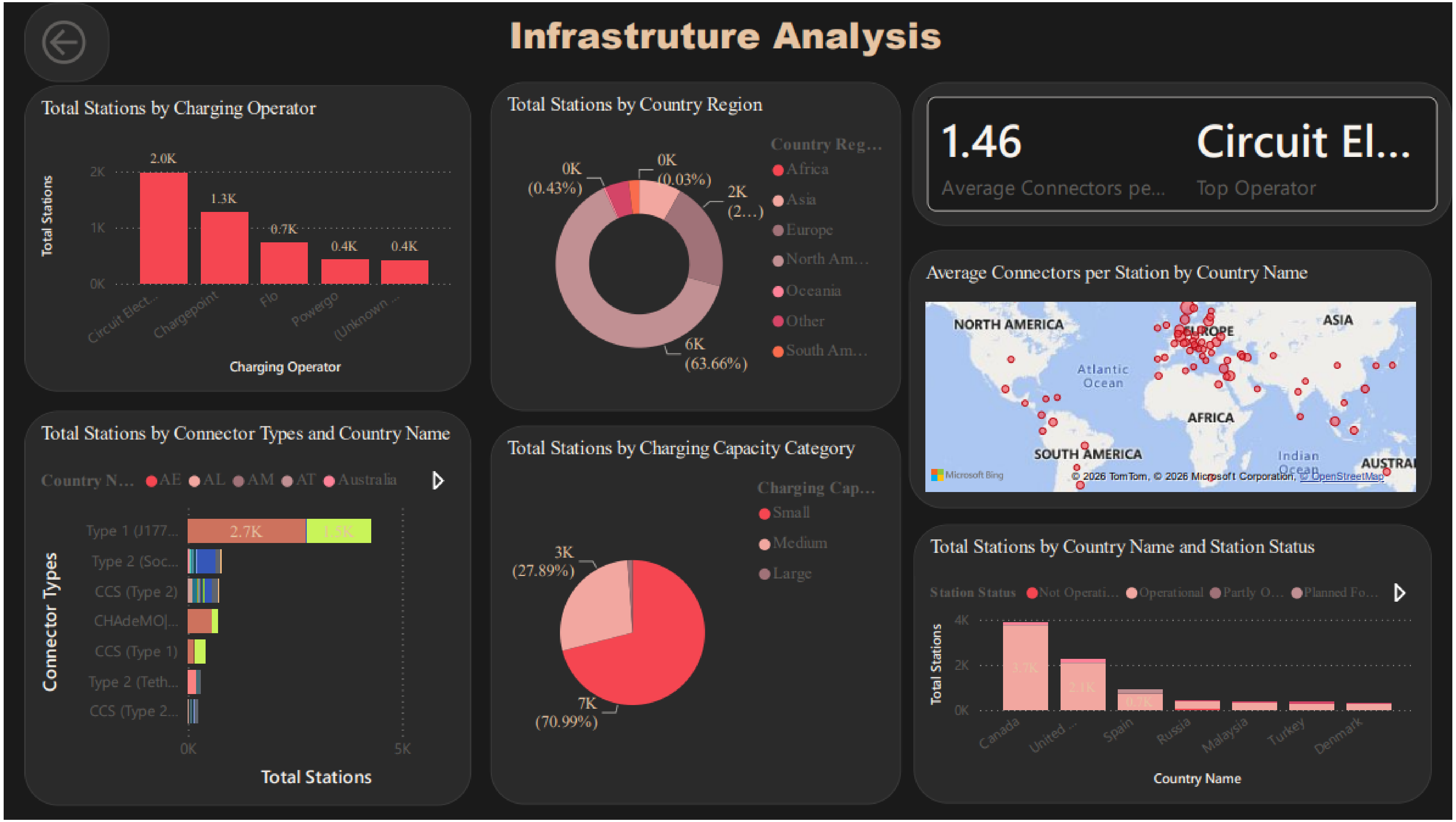
Operator & Regional Findings

- Average Connectors per Station: 1.46
- Top Operator: Circuit Electrique
- Circuit Electrique: about 2.0K stations
- Chargepoint: about 1.3K stations
- FLO: about 0.7K stations
- North America is the dominant region

Capacity & Connector Findings

- Small stations: about 70.99%
- Medium stations: about 27.89%
- Large stations: very small share
- Type 1 (J1772) is prominent in connector analysis
- Country/status charts enable detailed comparison

Page 2: Infrastructure Analysis



Page 3: Station Details & Drillthrough

From aggregate results to station-level investigation

Station-Level Table

- Station ID
- Country Name
- Charging Operator
- Total Connectors
- Operational Flag
- Detailed station records
- Charging capacity distribution

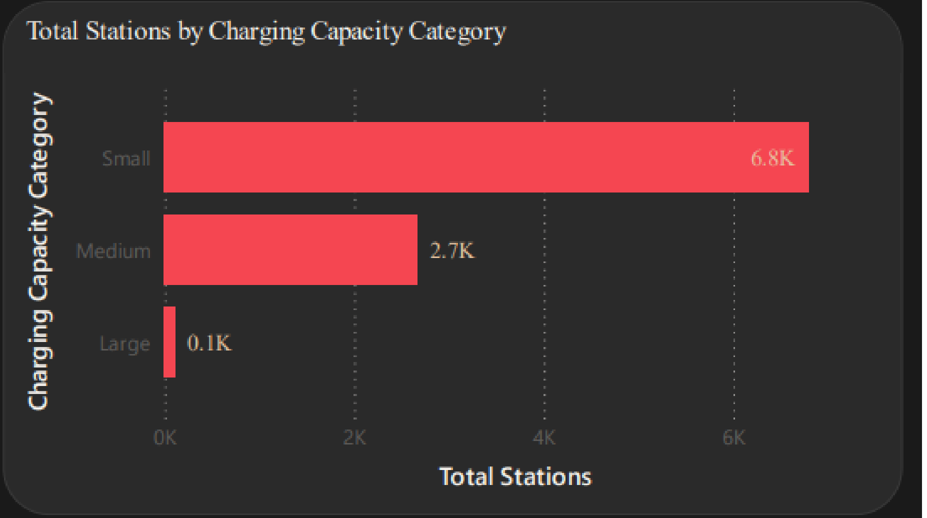
Regional Summary

- North America: 6,092 total; 5,828 operational; 264 non-operational
- Europe: 2,017 total; 1,792 operational; 225 non-operational
- Asia: 782 total; 701 operational; 81 non-operational
- South America: 199 total; 173 operational; 26 non-operational

Page 3: Station Details & Drillthrough

Station Details & Drillthrough Analysis

Station ID	Country Name	Charging Operator	Total Connectors	Operational Flag
374562	Canada	Flo	2	Yes
374563	United States	Opconnect	1	Yes
374564	Canada	Unknown Operator	1	Yes
374565	Canada	Flo	2	Yes
374567	United States	Blink Charging	1	Yes
374569	Canada	Circuit Electrique	1	Yes
374573	Canada	Chargepoint	1	Yes
374575	United States	Chargepoint	1	Yes
374577	Canada	Flo	2	Yes
374579	United States	Blink Charging	1	Yes
374580	Canada	Circuit Electrique	1	Yes
374581	Canada	Circuit Electrique	1	Yes
374582	United States	Blink Charging	1	Yes
374583	Canada	Flo	1	Yes
374585	Canada	Circuit Electrique	1	Yes
374586	Canada	Circuit Electrique	1	Yes
374589	Canada	Circuit Electrique	1	Yes
374592	United States	Chargepoint	2	Yes
374594	United States	Chargepoint	2	Yes
374595	Canada	Flo	1	Yes
374597	Canada	Flo	1	Yes
374598	Canada	Incharge (Us)	1	Yes
374599	Canada	Circuit Electrique	1	Yes
Total			**13974**	



Country Region	Total Stations	Operational Stations	Non Operational Stations
Africa	3	3	
Asia	782	701	81
Europe	2017	1792	225
North America	6092	5828	264
Oceania	41	41	
Other	436	406	30
South America	199	173	26
Total	**9570**	**8944**	**626**

Key Insights

Major findings from the completed dashboard

1	**High operational availability**	93.46% of stations are reported as operational.
2	**North America dominates**	6,092 stations are shown in the regional summary.
3	**Canada leads**	About 3.8K stations in the Top 5 country chart.
4	**Operator concentration**	Circuit Electrique is the leading displayed operator at about 2.0K stations.
5	**Small stations dominate**	Small capacity represents about 70.99% of stations.

Data Consistency Observation

A validation point identified from the completed dashboard

Executive Overview

- Total Stations: 10K
- Operational Stations: 9K
- Non Operational Stations: 626
- Operational %: 93.46%
- Total Connectors: 14K

Station Details Regional Table

- Regional total: 9,570 stations
- Operational: 8,944
- Non Operational: 626
- The non-operational count matches the overview.
- The total differs from the 10K headline, so filter/context consistency should be checked.

Business Value

How the dashboard supports analysis and decision-making

Infrastructure Planning

- Compare countries and regions
- Analyze capacity distribution
- Identify areas for further investigation

Operator Analysis

- Compare operator station counts
- Understand operator concentration
- Identify unknown operator data

Operational Analysis

- Monitor operational status
- Compare regional performance
- Investigate station-level records

Future Scope

Potential extensions beyond the current dashboard

Advanced Analytics

- Real-time charging station APIs
- Charging power and AC/DC analysis
- EV registrations and population integration
- Predictive charging-demand analysis
- Geographic optimization for new locations

Dashboard Enhancements

- Automated refresh
- Station utilization and availability trends
- Distance-to-nearest-station analysis
- Operator reliability and growth analysis
- Infrastructure expansion recommendations

Conclusion

Summary of the completed project

- The project transforms raw EV charging station data into an interactive Power BI intelligence dashboard.
- Power Query and M Language clean, standardize and transform the dataset.
- DAX calculated columns and measures create meaningful analytical indicators.
- The three pages provide executive overview, infrastructure comparison and station-level investigation.
- The dashboard highlights high operational availability, strong North American concentration, Canada as the leading displayed country, Circuit Electrique as the leading displayed operator, and dominance of small stations.
- The solution provides a foundation for future real-time, predictive and geographic EV infrastructure analytics.

Thank You